

Comparison of Precipitation in Seattle and New York

Objective

The analysis focuses on comparing precipitation patterns between Seattle and New York. Both visualizations and statistical methods were applied to identify similarities and differences.

Data Preparation

Datasets for Seattle and New York were concatenated into a combined dataset. Columns were standardized ('PRCP' to 'precipitation' and 'DATE' to 'Date'), and a 'city' column was added for identification. Dates were converted to datetime format.

Visualization

Several plots were created: - Line plots showing daily precipitation trends. - KDE plots (e.g., January) to show precipitation distribution. - Bar charts of mean daily precipitation. These visuals helped highlight differences: Seattle tends to have frequent light rain, while New York experiences fewer but heavier rainfall events.

Numerical Summaries

Descriptive statistics (mean, std, quartiles, min, max) were computed for precipitation by city. This quantified Seattle's more consistent light rainfall and New York's more variable, heavier events.

Statistical Testing: T-Test

Two-sample t-tests were performed month by month to compare daily precipitation means. Significant differences ($p < 0.05$) indicate that average rainfall varies between the two cities in certain months.

Statistical Testing: Proportion of Rainy Days

A two-proportion z-test was applied to compare the proportion of days with precipitation (>0 inches) in each city. Results confirmed Seattle has a higher proportion of rainy days, while New York tends to have fewer but more intense events.

Conclusion

The combined results show that Seattle experiences light but frequent rainfall, whereas New York has more variability and heavier rain events on days it rains. Statistical testing supports these findings, demonstrating significant differences in both mean precipitation and the proportion of rainy days across months.